

MeSHClass-ES and AnatEM-ES

Open Resources for Spanish Biomedical NLP

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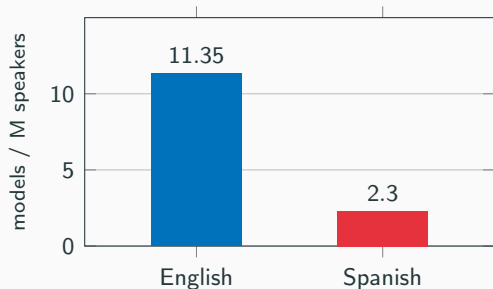
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Agenda

1. Motivation
2. The Resources
3. The Benchmark
4. Conclusions

Motivation

Spanish biomedical NLP is under-resourced



HuggingFace models per million speakers

- English has domain models: BioBERT, PubMedBERT, BioGPT
- Spanish: a handful of initiatives
- The gap is largest in the biomedical domain

Two open questions

- Q1 Which Spanish or multilingual **encoder** works best for biomedical tasks, and how much does domain pretraining matter?
- Q2 Can open-weight **decoder LLMs** substitute for fine-tuned encoders without task-specific training?

We study both on two tasks: **MeSH classification** and **anatomical NER**

Two tasks, no open Spanish resource

Broad-topic MeSH classification

Document-level labelling into MeSH categories

No open Spanish corpus at PubMed scale

Fine-grained anatomical NER

Token-level tagging of 12 anatomical types

No native Spanish corpus at this granularity

Spanish datasets exist for drugs, diseases, and symptoms, but not anatomy

We build a Spanish benchmark by translating the English AnatEM dataset

- **Spanish biomedical models are scarce:** a few clinical-focused encoders (RoBERTa-Bio-Clinical-ES, RigoBERTa-Clinical, SciBETO), no broad biomedical suite as in English
- **MeSH indexing and most benchmarks are English-only:** BioASQ, DeepMeSH, BLURB. No open Spanish resource for either of our two tasks

- 1 **Two Spanish biomedical corpora**, each translated with an approach matched to its task, released bilingual
- 2 An **encoder benchmark** of 9 models: general-domain Spanish, domain-specific, and multilingual
- 3 A **decoder evaluation** of 4 open-weight LLMs (7–9B) in zero-shot, few-shot, retrieval, and QLoRA settings

The Resources

Two corpora

MeSHClass-ES

- 44,604 PubMed abstracts
- 29,063 with major-topic MeSH labels
- 14 Level-1 categories, multi-label
- Translated with Opus-MT

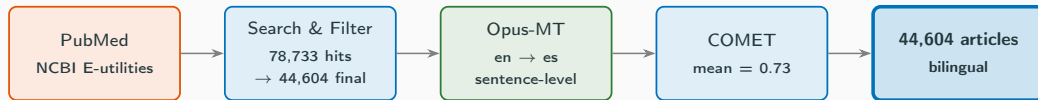
AnatEM-ES

- 1,212 documents
- 13,849 entity mentions
- 12 anatomical types, BIO tagging
- Translated with GPT-4o

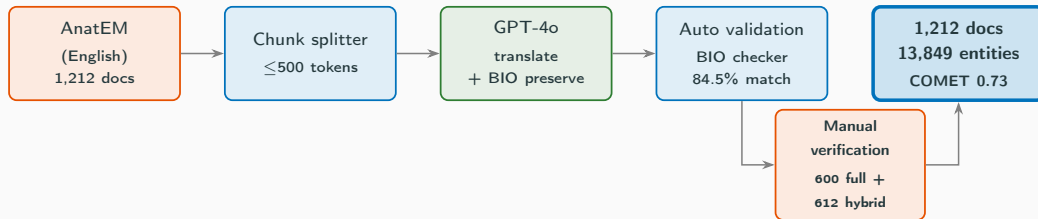
Both are COMET-assessed, bilingual, and released on the HuggingFace Hub

Construction pipelines

MeSHClass-ES



AnatEM-ES



Translation quality: a domain-level ceiling

- Both corpora reach mean **COMET 0.73**
- Opus-MT and GPT-4o converge: the ceiling is the **domain**, not the system
- AnatEM-ES entities human-verified (600 full + 612 hybrid)
- Bilingual release enables re-translation as MT improves

The Benchmark

9 encoders, fine-tuned:

- General Spanish: BETO, BERTIN, RigoBERTa-2.0
- Domain: SciBETO-base/large, RigoBERTa-Clinical, RoBERTa-Bio-Clinical-ES
- Multilingual: XLM-RoBERTa base / large

4 decoders (7–9B):

- Gemma-2-9B, Llama-3.1-8B, Qwen2.5-7B, Mistral-7B
- Zero-shot, few-shot, k-NN retrieval, QLoRA

Tasks

- Classification: 14 categories, multi-label; micro / macro-F1
- NER: 12 types, AnatEM splits; entity-level F1 (seqeval)

Model	Mi-F1	Ma-F1
RigoBERTa-2.0	.692	.584
SciBETO-large	.692	.577
XLM-R (large)	.688	.578
RigoBERTa-Clinical	.681	.580
BERTIN	.679	.573
SciBETO-base	.678	.561
BETO	.675	.553
RoBERTa-Bio-Clin.	.675	.557
XLM-R (base)	.666	.556

- All 9 models fall in a 3-point band (.67–.69)
- RigoBERTa-2.0 leads, tied with SciBETO-large
- General-domain BETO / BERTIN sit mid-table

Classification — decoders

Setting	Model	Mi-P	Mi-R	Mi-F1	Ma-F1
<i>Zero-shot</i>					
	Gemma-2-9B	.352	.654	.458	.282
	Llama-3.1-8B	.398	.459	.426	.178
	Qwen2.5-7B	.401	.438	.418	.216
	Mistral-7B	.259	.564	.355	.190
<i>Few-shot (3 ex.)</i>					
	Gemma-2-9B	.325	.615	.426	.289
	Llama-3.1-8B	.317	.497	.387	.217
	Qwen2.5-7B	.292	.412	.342	.234
	Mistral-7B	.284	.591	.384	.244
<i>QLoRA</i>					
	Gemma-2-9B	.619	.597	.608	.500
	Mistral-7B	.611	.475	.535	.450
	Llama-3.1-8B	.564	.343	.427	.367
	Qwen2.5-7B	.593	.320	.416	.358

- QLoRA Gemma-2-9B: **.608**, 88% of the best encoder
- Few-shot lowers micro-F1 but raises macro-F1
- Model choice matters under QLoRA

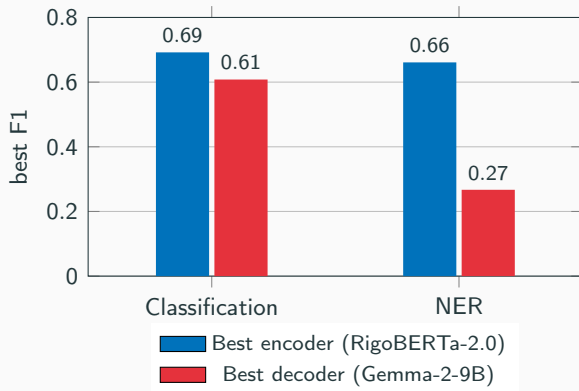
Model	P	R	F1
RigoBERTa-2.0	.661	.661	.661
RigoBERTa-Clinical	.638	.664	.651
XLM-R (large)	.640	.655	.647
SciBETO-large	.638	.648	.643
RoBERTa-Bio-Clin.	.644	.622	.633
BETO	.624	.630	.627
BERTIN	.620	.622	.621
XLM-R (base)	.616	.621	.618
SciBETO-base	.612	.623	.618

- Spread is 4 points, wider than classification
- RigoBERTa-2.0 leads both tasks
- Mid-table order changes:
task-specific evaluation matters

Setting	Model	P	R	F1
<i>Zero-shot</i>				
	Gemma-2-9B	.243	.295	.267
	Llama-3.1-8B	.117	.211	.151
	Qwen2.5-7B	.132	.126	.129
	Mistral-7B	.115	.129	.122
<i>k-NN few-shot (k=5)</i>				
	Gemma-2-9B	.228	.012	.024
	Llama-3.1-8B	.266	.012	.023
<i>QLoRA</i>				
	Gemma-2-9B	.002	.006	.003
	Mistral-7B	.005	.004	.004

- Best decoder (Gemma .27): 40% of the best encoder
- Failures are **structural**: all-O or malformed BIO under QLoRA, pattern-copying under k-NN
- Model choice does not rescue the formulation

The encoder-decoder gap depends on the task



- Classification: decoder reaches 88% of the encoder
- NER: decoders stay at or below 40%
- LLMs handle topic labelling, not span detection

Conclusions

1. Two open Spanish biomedical corpora: **MeSHClass-ES** and **AnatEM-ES**, bilingual and COMET-assessed
2. **RigoBERTa-2.0** leads both tasks; capacity and domain pretraining both help, more on NER
3. **XLNet-RoBERTa-large** is a strong multilingual baseline
4. Decoders do not replace encoders yet: 88% on classification, but $\leq 40\%$ on NER

Thanks for your attention!

Corpora, translation pipelines, and evaluation code are openly released



MeSHClass-ES

HuggingFace



AnatEM-ES

HuggingFace



Code

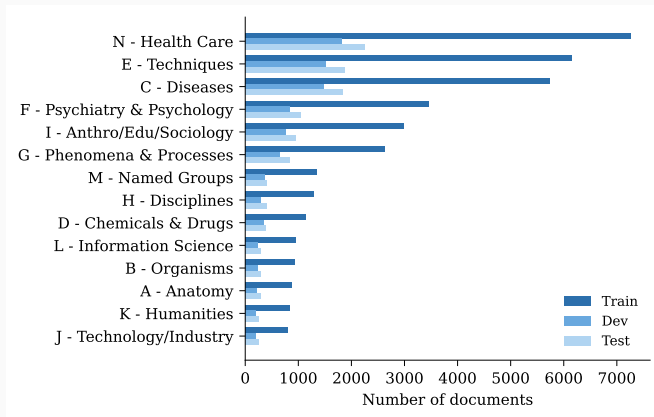
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MeSHClass-ES category distribution

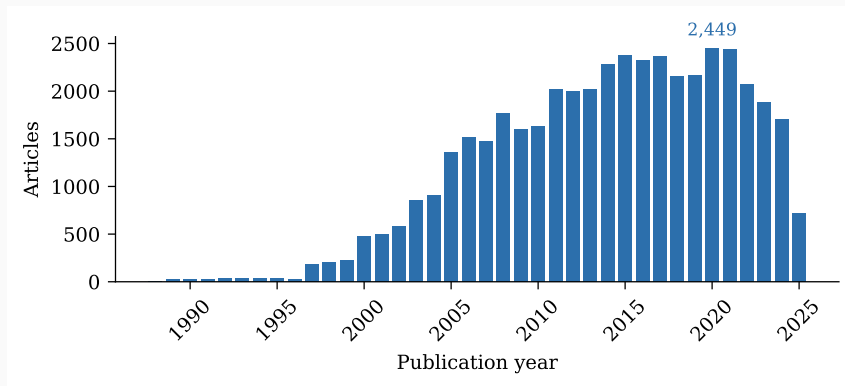


MeSH Level-1 category distribution (train/dev/test). Health Care (N) and Techniques (E) dominate; a 9:1 imbalance between the most and least frequent categories.

AnatEM-ES entity distribution

Entity Type	Train	Dev	Test
Anatomical_system	67	7	39
Cancer	1,469	513	1,129
Cell	2,459	666	1,447
Cellular_component	413	138	287
Developing_anat_str.	45	20	35
Immaterial_anat_ent.	116	50	97
Multi-tissue_str.	884	274	559
Organ	413	146	306
Organism_subdiv.	166	62	109
Organism_subst.	338	102	244
Pathological_form.	182	63	151
Tissue	464	129	260
Total	7,016	2,170	4,663

Temporal distribution of MeSHClass-ES



Publication year distribution (1988–2025)